



*Figure 9-4: A selection of branches from the decision tree*

We've successfully created a decision tree that enables us to predict happiness levels with relatively low error. You can examine the output to see the relative determinants of happiness, and the happiness levels associated with each branch.

There is more exploring we can do with decision trees and our dataset. For example, we can try to run the same code but with a different or larger set of variables. We can also create a tree with a different maximum depth. Here is an example of running the code with a different variable list and depth:

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```

variables = ['sclmeet', 'rlgdgr', 'hhmb', 'netusoft', 'agea', 'eduyrs', 'health']
outcome_variable = 'happy'
maxdepth = 3
print(getsplit(0, ess, variables, outcome_variable))

```

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When we run it with these parameters, we find a very different decision tree. You can see the output here:

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```

[['health', -inf, 2.0, [['sclmeet', -inf, 4.0, [['health', -inf, 1.0, [['rlgdgr',
-inf, 9.0, 7.9919636617749825], ['rlgdgr', 9.0, inf, 8.713414634146341]]],
['health', 1.0, inf, [['netusoft', -inf, 4.0, 7.195121951219512], ['netusoft',
4.0, inf, 7.565659008464329]]]]], ['sclmeet', 4.0, inf, [['eduyrs', -inf, 25.0,
[['eduyrs', -inf, 8.0, 7.9411764705882355], ['eduyrs', 8.0, inf,
7.999169779991698]]], ['eduyrs', 25.0, inf, [['hhmb', -inf, 1.0,
7.297872340425532], ['hhmb', 1.0, inf, 7.9603174603174605]]]]]]], ['health',
2.0, inf, [['sclmeet', -inf, 3.0, [['health', -inf, 3.0, [['sclmeet', -inf, 2.0,
6.049427365883062], ['sclmeet', 2.0, inf, 6.70435393258427]]], ['health', 3.0,
inf, [['sclmeet', -inf, 1.0, 4.135036496350365], ['sclmeet', 1.0, inf,
5.407051282051282]]]]], ['sclmeet', 3.0, inf, [['health', -inf, 4.0, [['rlgdgr',
-inf, 9.0, 6.992227707173616], ['rlgdgr', 9.0, inf, 7.434662998624484]]],

```

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